ITES Case Study — Fraud Detection (Option B)

Task 1: Problem Type

PayPal fraud detection-this is a binary classification problem where the model predicts whether a transaction is fraud or not fraud. The challenge is class imbalance, since fraud cases are rare compared to normal transactions. False negative is so risky. The system primarily uses supervised learning, as it relies on labeled historical transaction data (fraud vs non-fraud). However, in real-world systems such as PayPal, unsupervised methods like anomaly detection are also used to detect new and unseen fraud patterns.

Task 2: Data Requirements

1. transaction_id – unique identifier for each transaction.

2. amount – large or unusual amounts may indicate fraud.

3. timestamp – unusual times may signal suspicious behavior.

4. sender_id – helps track user behavior patterns.

5. receiver_id – identifies suspicious recipient accounts.

6. device_info – unknown devices may indicate fraud.

7. location – unusual geographic locations may be risky.

8. transaction_frequency – sudden spikes may signal fraud.

9. account_age – newer accounts are often riskier.

10. transaction_type – some types are more vulnerable to fraud.

11. user_behaviour_pattern-deviation from normal linear motion.

Task 3: Evaluation Metric

F1-score is the best metric because it balances precision and recall. High precision reduces false alarms, while high recall ensures fraud cases are not missed. In fraud detection, missing fraud (low recall) can be very costly.

Task 4: Bias and Fairness Risks

The model may unfairly flag users from rural areas or certain regions due to limited training data. It may also discriminate based on income levels or device types. This could lead to unfair blocking of legitimate transactions. Old phone users may be flagged.

Task 5: Privacy Concerns

The system uses sensitive data like financial transactions, location, and device information. If breached, attackers could steal money or track users. Strong encryption and data minimization are necessary.

Task 6: Recommendation

The system should be deployed with guardrails. These include human review for flagged transactions, regular bias audits, and anonymization of user data. Users should also have the ability to appeal decisions.

Task 7: Reflection Questions

1. Yes, in countries with strict laws like GDPR, privacy concerns would be stronger and require stricter controls.

2. It is possible but less effective. Without personal data, the system loses accuracy and context.

3. Responsibility should be shared between developers and the company, not the algorithm itself.